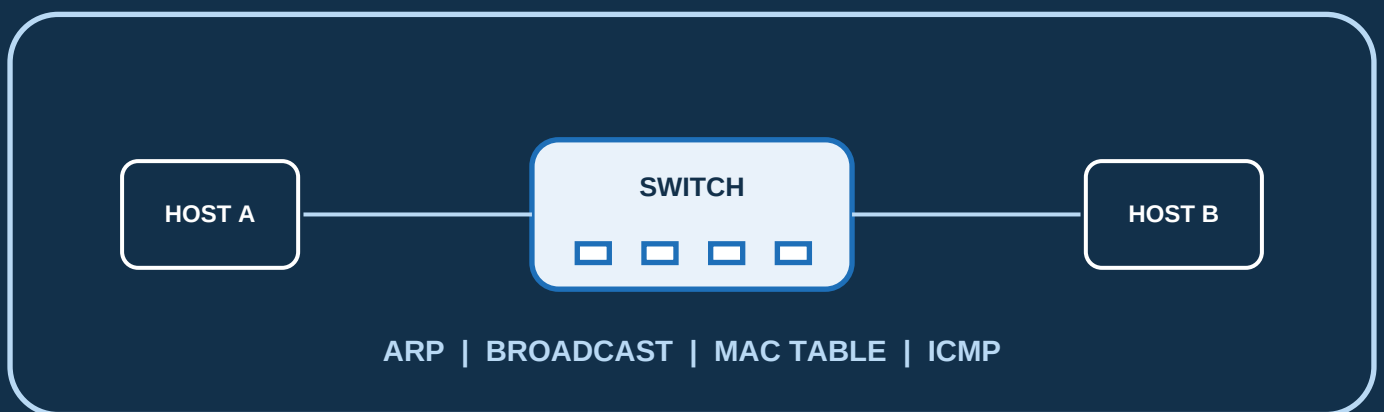


CCNA 200-301 VISUAL FLASHCARDS

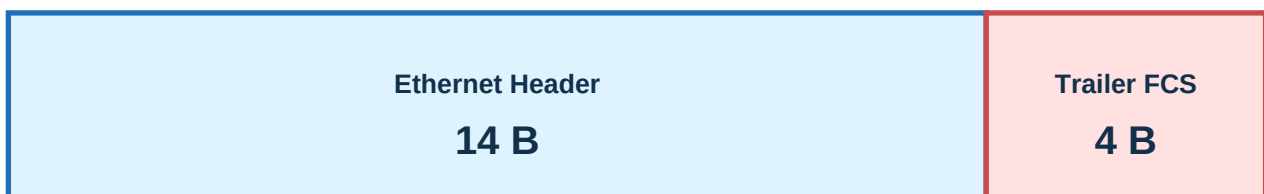
DAY 06

Ethernet LAN Switching Part 2

17 original flashcards - visualized and explained



What is the size of the Ethernet header + trailer? (no Preamble + SFD)



14-byte header + 4-byte FCS trailer = 18 bytes

ANSWER

18 bytes

MEMORY CUE

14-byte Ethernet header + 4-byte FCS trailer = 18 bytes.

What is the minimum size of an Ethernet frame? (Header + Payload + Trailer)



Minimum Ethernet frame = $14 + 46 + 4 = 64$ bytes

ANSWER

64 bytes

MEMORY CUE

Minimum frame size is 64 bytes, excluding Preamble and SFD.

What is the minimum size of an Ethernet payload?

Minimum Payload

46 B

Ethernet pads small payloads so the data field reaches at least 46 bytes

ANSWER

46 bytes

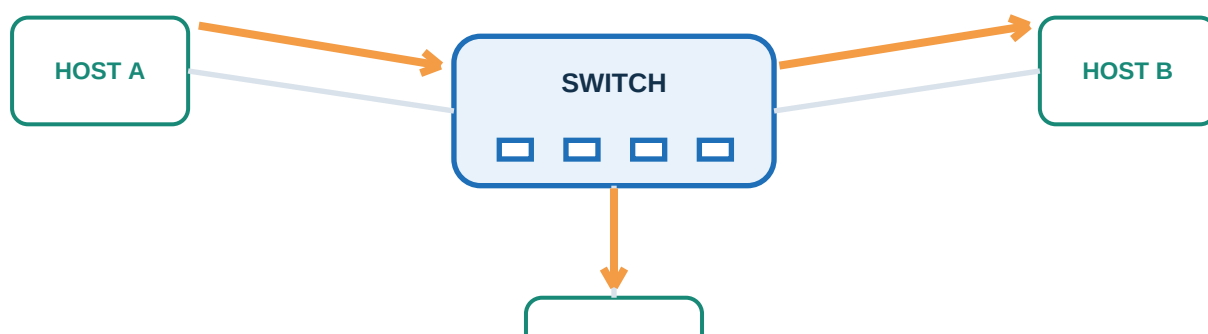
MEMORY CUE

Small payloads are padded to at least 46 bytes.

What protocol is used to discover the Layer 2 address of another device?

1. ARP Request: broadcast

2. ARP Reply: unicast



ANSWER

ARP (Address Resolution Protocol)

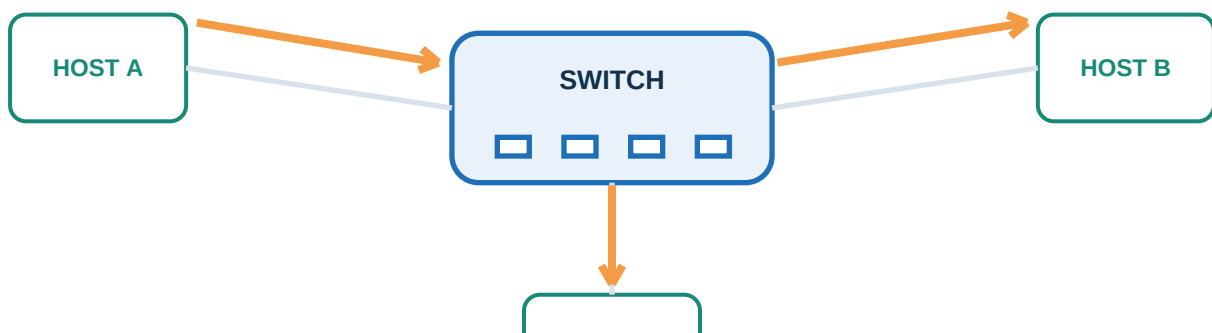
MEMORY CUE

ARP maps a known IPv4 address to a Layer 2 MAC address.

What two messages are used in ARP?

1. ARP Request: broadcast

2. ARP Reply: unicast



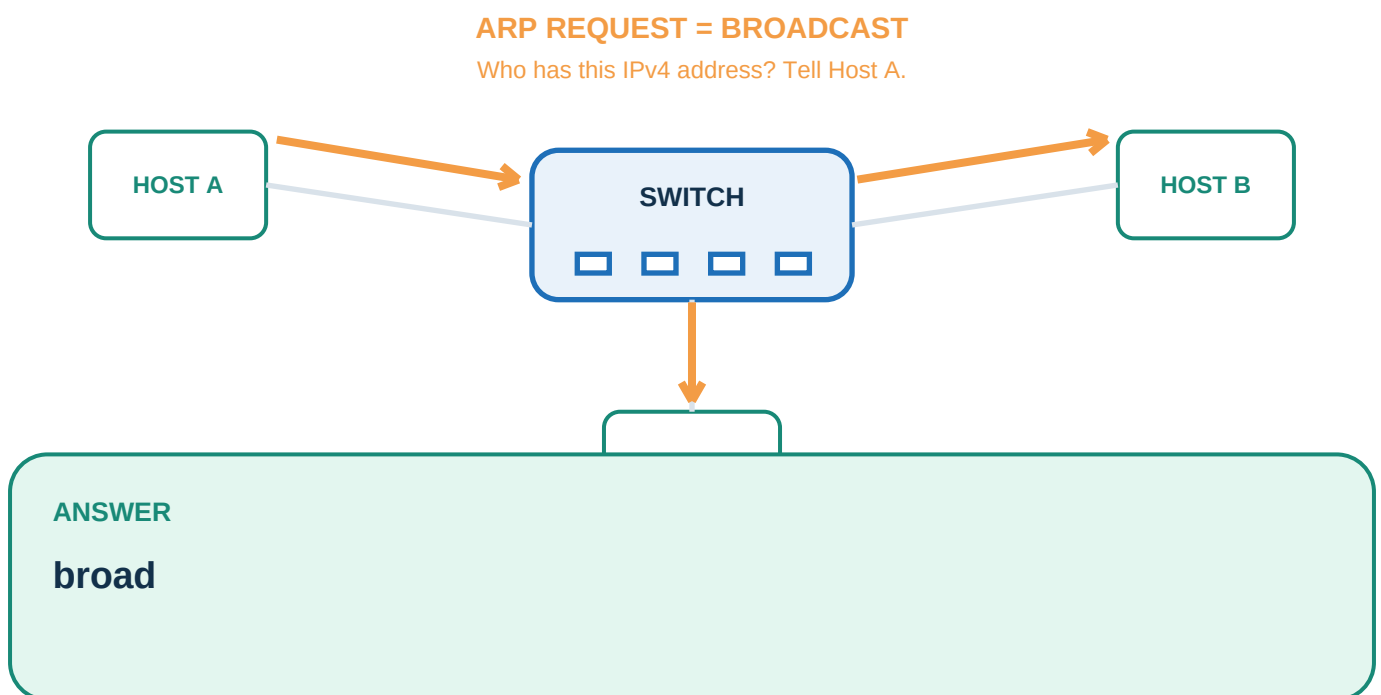
ANSWER

ARP Request
ARP Reply

MEMORY CUE

Request asks who owns the IP; Reply provides the MAC.

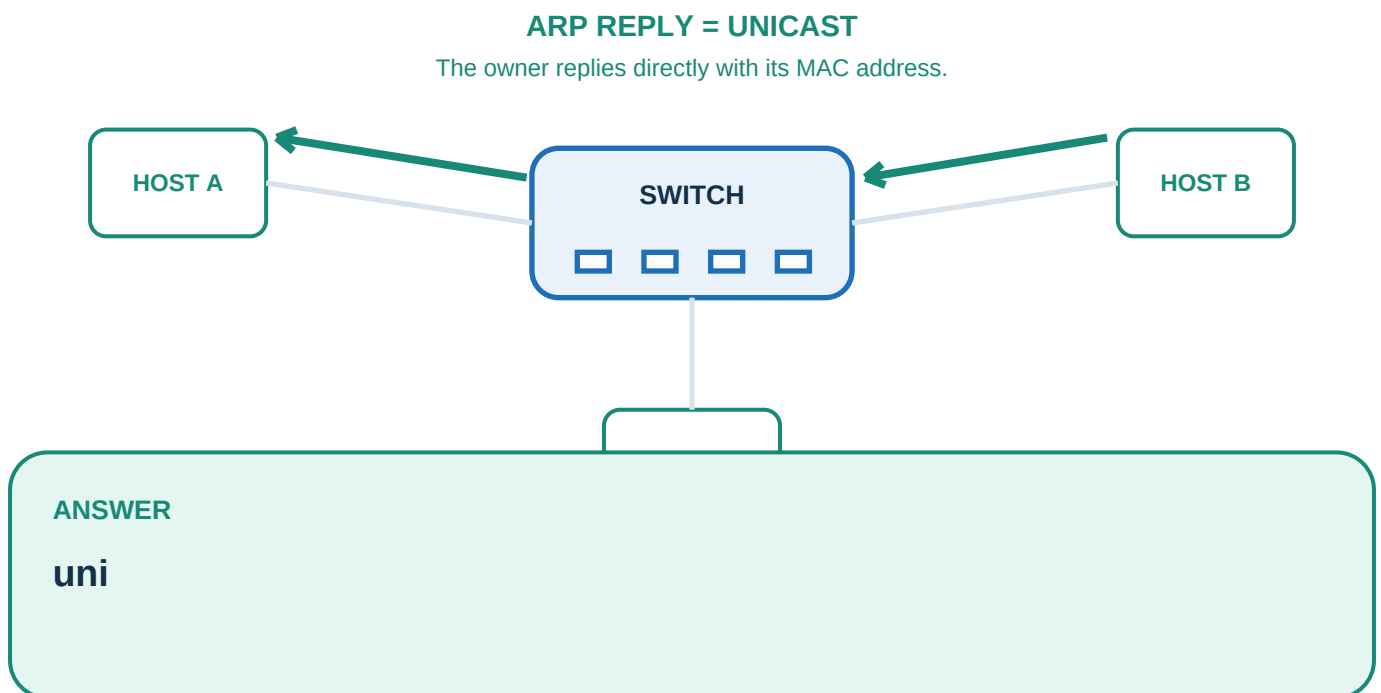
The ARP Request message is [...]cast.



MEMORY CUE

ARP Request must reach everyone locally, so it is broadcast.

The ARP Reply message is [...]cast.



MEMORY CUE

ARP Reply goes directly back to the requester, so it is unicast.

What is the broadcast MAC address?

FFFF.FFFF.FFFF

ALL 48 BITS ARE 1

A switch floods a Layer 2 broadcast out all other ports in the VLAN.

ANSWER

FFFF.FFFF.FFFF

MEMORY CUE

FFFF.FFFF.FFFF is the Layer 2 broadcast address.

What is the command to view the ARP table on a Windows, macOS, or Linux computer?

```
> arp -a
```

Host OS command

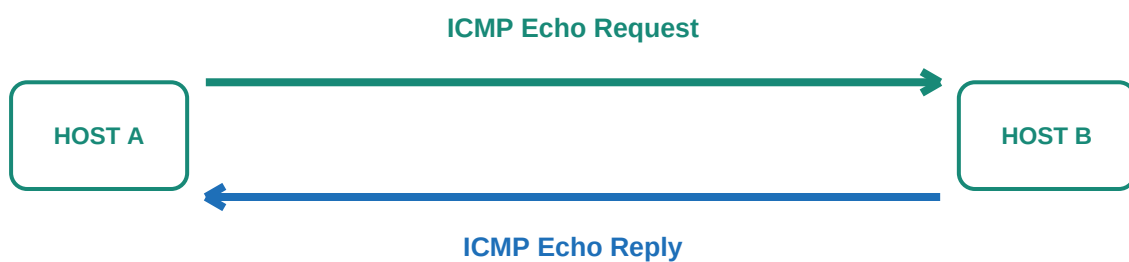
ANSWER

arp -a

MEMORY CUE

On common host OSes, arp -a displays cached IP-to-MAC mappings.

What two messages are used in ping?



ping tests reachability using ICMP echo messages

ANSWER

ICMP Echo Request
ICMP Echo Reply

MEMORY CUE

ping uses ICMP Echo Request and ICMP Echo Reply.

The process of addresses being cleared from the MAC address table after 5 minutes of inactivity is known as [...]

MAC ADDRESS TABLE

VLAN	MAC Address	Port
1	00AA.11BB.22CC	Gi0/1
1	00DD.11EE.22FF	Gi0/2

**5 min inactivity****Entry removed = aging****ANSWER****aging****MEMORY CUE**

Dynamic switch MAC entries age out after 5 minutes of inactivity in this course context.

Clear all dynamic MAC addresses from the MAC Address Table:
SW1# [...]

```
SW1# clear mac address-table dynamic
```

Cisco IOS command

ANSWER

```
clear mac address-table dynamic
```

MEMORY CUE

This removes all dynamically learned MAC entries from the switch table.

View the MAC Address Table: SW1# [...]

```
SW1# show mac address-table
```

Cisco IOS command

ANSWER

```
show mac address-table
```

MEMORY CUE

show mac address-table is the key verification command.

Clear all entries for a specific dynamic MAC address from the MAC Address Table:

SW1# [...]

```
SW1# clear mac address-table dynamic address mac-address
```

Cisco IOS command

ANSWER

```
clear mac address-table dynamic address mac-address
```

MEMORY CUE

Use the address keyword when clearing one learned MAC entry.

Clear all dynamic MAC addresses on a specific interface from the MAC Address Table:

SW1# [...]

```
SW1# clear mac address-table dynamic interface interface-id
```

Cisco IOS command

ANSWER

```
clear mac address-table dynamic interface interface-id
```

MEMORY CUE

Use the interface keyword to clear dynamic entries learned on one port.

What is the Ethernet Type code for an ARP packet?

ARP EtherType
0x0806

ANSWER

0x0806

MEMORY CUE

ARP is identified in Ethernet by EtherType 0x0806.

Command to show the ARP table on a Cisco device:

```
SW1# show arp
```

Cisco IOS command

ANSWER

```
show arp
```

MEMORY CUE

Cisco IOS: show arp displays the ARP cache.

Ethernet sizes

Header + trailer = 18 B; minimum payload = 46 B; minimum frame = 64 B.

ARP

ARP resolves IPv4 -> MAC. Request = broadcast; Reply = unicast.

Broadcast MAC

FFFF.FFFF.FFFF

Host ARP table

arp -a

Cisco ARP table

show arp

Ping

ICMP Echo Request + ICMP Echo Reply

MAC aging

Dynamic entries age out after 5 minutes of inactivity in this deck.

Show MAC table

show mac address-table

Clear all dynamic

clear mac address-table dynamic

Clear one MAC

clear mac address-table dynamic address mac-address

Clear one interface

clear mac address-table dynamic interface interface-id

ARP EtherType

0x0806